FINITELY MANY ADMISSIBLE BLOWUPS ARE DOMINATED BY ONE

THE STACKS PROJECT AUTHORS

Editorial note. All delivered mathematical text is the author's own, copied from the pinned Stacks Project chapter file divisors.tex; no statement or proof step was rewritten, reordered or removed.

Editorial note. Carrier window: statement and proof both lie inside the window 8000-10000 granted to this lane by CONTROL 8.83.

Editorial note. Part (b) of the statement rests on the same cited two-ideals lemma that gives part (a); the proof states the construction in place and delegates the comparison to that lemma, which is recorded here for the reader.

Editorial note. The single reference of the counted spans is recorded as a non-load-bearing navigation pointer; the argument that consumes it stays inside the counted proof span.

Editorial note. Preamble: sources/stacks-divisors/retained-preamble.tex (88 lines, verbatim present).

Allowed standard prerequisites (editorial).

- Comparison of blowing up in a product of ideals. Lemma lemma-blowing-up-two-ideals (8588-8595), used at proof lines 9165-9167. Gives the factorization X' -> X_i -> X.
- Description of an admissible blowup by a finite type ideal. used at proof lines 9160-9164. Supplies the ideals I_i.

Source reference. Original lemma 080A: focused-seeds/AGP/sources/stacks-snapshot-20261011/tags.txt, source lines 10175–10175. Navigation pointer into the Stacks Project; the standard fact it supplies is applied inside the counted proof span and the argument resting on it stays there.

Lemma 0.2. *Let X be a quasi-compact and quasi-separated scheme. Let $U \subset X$ be a quasi-compact open subscheme. Let $b_i : X_i \to X$, $i = 1, \ldots, n$ be U-admissible blowups. There exists a U-admissible blowup $b : X' \to X$ such that (a) b factors as $X' \to X_i \to X$ for $i = 1, \ldots, n$ and (b) each of the morphisms $X' \to X_i$ is a U-admissible blowup.*

Proof. Let $\mathcal{I}_i \subset \mathcal{O}_X$ be the finite type quasi-coherent sheaf of ideals such that $V(\mathcal{I}_i)$ is disjoint from U and such that X_i is isomorphic to the blowup of X in $\mathcal{I}_i$. Set $\mathcal{I} = \mathcal{I}_1 \cdot \ldots \cdot \mathcal{I}_n$ and let X' be the blowup of X in $\mathcal{I}$. Then $X' \to X$ factors through b_i by Lemma 080A. □

Date: stacks-project:divisors.tex:local-snapshot-20261012; GNU FDL 1.2 or later.